ETF-TPT: Equiangular Tight Frame Guided Test-Time Prompt Tuning for Calibrated Zero-Shot Generalization in Vision–Language Models

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Abstract

Vision–language models (VLMs) such as CLIP achieve strong zero-shot generalization by aligning image and text in a shared embedding space. Test-Time Prompt Tuning (TPT) further improves zero-shot accuracy by optimizing learnable prompt tokens per test image through entropy minimization, but *systematically* miscalibrates the model: predicted probabilities become persistently overconfident and no longer reflect the true likelihood of correctness, an unacceptable failure mode in safety-critical deployments. Existing calibration-aware variants disperse class text features geometrically, yet they suffer from pairwise feature collapse, geometrically infeasible targets, or unstable greedy gradients. We propose ETF-TPT, which regularizes the per-test-image prompt geometry toward an *Equiangular Tight Frame* (ETF)—the maximally uniform arrangement of N unit vectors in $\mathbb{R}^D$ permitted by frame theory. We show that the ETF target subsumes prior centroid-, orthogonality-, and angular-margin-based regularizers as special or degenerate cases. While direct ETF regularization sharply improves calibration, it homogenizes pre-trained semantic relations and degrades accuracy. To resolve this tension we introduce **Semantic-Aware ETF** ($\mathcal{L}_{\text{S-ETF}}$), a margin-constrained, ReLU-gated relative target that enforces near-uniform angular dispersion *while* preserving the semantic topology of the underlying text encoder. The resulting regularizer is bounded below by the Welch limit, adds no inference-time cost, and exposes only two stable hyperparameters. Across 11 classification benchmarks and two CLIP backbones (ViT-B/16 and RN50), S-ETF-TPT attains state-of-the-art calibration (average ECE of 1.72 on ViT-B/16 vs. 2.61 for the strongest prior method) while matching or improving zero-shot accuracy.

1 Introduction

Foundation vision–language models (VLMs) such as CLIP [Radford et al., 2021], ALIGN [Jia et al., 2021], and FILIP [Yao et al., 2021] have reshaped zero-shot recognition by contrastively aligning images and text in a shared multimodal embedding space trained on hundreds of millions of web image–caption pairs. Given a hand-crafted textual template (e.g., `"a photo of a [class]"`), they classify novel categories *without any labeled target data* by

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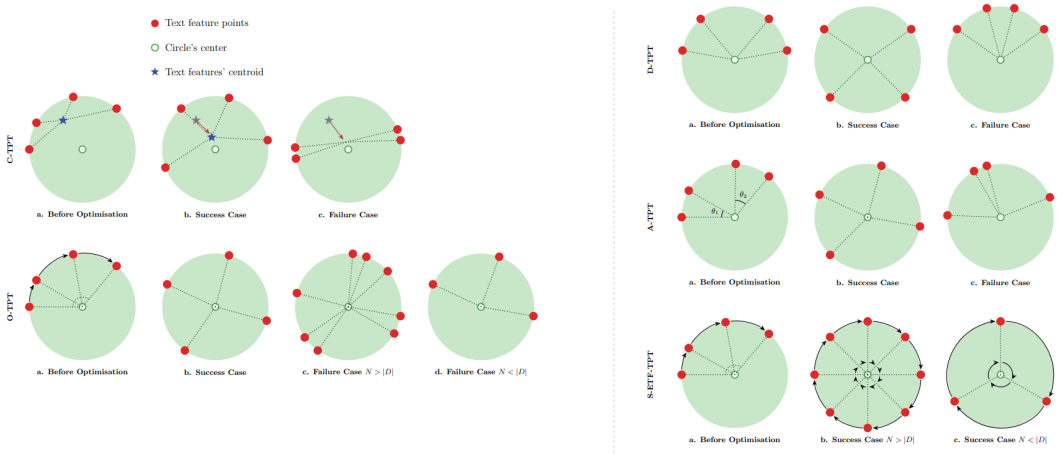


Figure 1: Geometry of class-text features after test-time prompt tuning under different dispersion objectives. Existing methods (C-TPT, O-TPT, D-TPT, A-TPT) admit failure cases in which features remain clustered or collapse into a shared subspace when $N > D$ or $N < D$. The proposed Semantic-ETF (S-ETF-TPT, bottom right) is the only objective whose Gram matrix attains the Welch-bound arrangement in both dimensionality regimes.

comparing image embeddings against per-class text embeddings. The quality of zero-shot inference therefore hinges on the prompt: small wording changes can swing accuracy by several points [Shu et al., 2022, Zhou et al., 2022b], and engineering templates by hand is brittle and dataset-specific.

From prompt engineering to test-time prompt tuning. Prompt tuning [Zhou et al., 2022b,a] replaces manual templates with continuous, learnable prompt tokens. As this requires labeled data and may overfit to a single domain, *Test-Time Prompt Tuning* (TPT) [Shu et al., 2022] adapts prompts *on the fly* for each unlabeled test image by minimizing the entropy of predictions averaged over augmented views. TPT consistently improves accuracy and has become a default backbone for zero-shot adaptation of VLMs [Zhou et al., 2022a, Sharifdeen et al., 2025, Yoon et al., 2024].

The hidden cost: severe miscalibration. TPT's accuracy gains conceal a fundamental reliability defect: the adapted prompts inflate logit magnitudes and produce *systematically overconfident* predictions whose probabilities no longer track the true likelihood of correctness [Guo et al., 2017, Murugesan et al., 2024, Yoon et al., 2024]. This is not cosmetic: in medical imaging, autonomous driving, and embodied agents [Wang et al., 2022, Ma et al., 2024, Liu et al., 2023, Dorbala et al., 2022, Khandelwal et al., 2022, Cui et al., 2024, Zhou et al., 2024], decisions are gated by predictive confidence; a model that is wrong yet confident is worse than one that abstains. Reliable uncertainty estimation is therefore not optional but a prerequisite for VLM deployment.

Why existing TPT-calibration methods fall short. Recent work tries to mitigate the overconfidence by *geometrically dispersing* the per-class text features so that inter-class similarities (and hence logit magnitudes) shrink. We identify four structural limitations shared by these methods, illustrated in fig. 1 and analyzed in section 2. **(i) Centroid-based dispersion** (C-TPT [Yoon et al., 2024]) maximizes the mean ℓ_2 distance of class features from their centroid, which leaves many pairs collapsed since pairwise angles are not constrained. **(ii) Orthogonality** (O-TPT [Sharifdeen et al., 2025]) targets pairwise $90°$ separation, which is geometrically *infeasible* whenever $N > D$ (the typical regime for ImageNet-21k or fine-grained datasets) and *suboptimal* when $N \leq D$ (the attainable limit is strictly larger than $90°$). **(iii) Greedy angular methods** such as A-TPT [Ahamed et al., 2025] maximize only the *minimum* pairwise angle, producing unstable gradients that allow non-minimal pairs to drift inward. **(iv) Dimensional entropy** (D-TPT) drives every

coordinate of every class vector toward $1/\sqrt{D}$, a single fixed point in the embedding space whose global minimum collapses all classes onto one direction. None of these objectives is the geometric optimum, and all distort the pre-trained semantic structure that makes zero-shot inference possible.

Our proposal: ETF as the principled geometric target. Frame theory provides a clean answer to the question *"what is the maximally separated configuration of N unit vectors in $\mathbb{R}^D$?"*: an *Equiangular Tight Frame* (ETF), in which *every* pair of vectors shares the same cosine similarity, and that similarity attains the Welch bound [Welch, 1974, Strohmer and Heath Jr., 2003]. ETFs unify the simplex limit ($N \leq D+1$) and the Welch arrangement ($N > D+1$) into a single principled target. We instantiate this insight in ETF-TPT, a drop-in regularizer that encourages the per-test-image text-feature Gram matrix toward the ETF target alongside the standard TPT entropy loss.

From vanilla ETF to Semantic-Aware ETF. A direct ETF regularizer dramatically reduces calibration error but collaterally degrades zero-shot accuracy. We trace this to a geometric pathology (section 3.3): when a class pair is already wider than the ETF target, the loss applies an *unwarranted contraction* that destroys useful pre-trained semantic relations. The embeddings of Car and Truck are, and should remain, closer to each other than to Cat; a uniform target homogenizes these meaningful asymmetries. We resolve the conflict with **Semantic-Aware ETF** (S-ETF-TPT), which (a) replaces the absolute uniform target with a relative *margin-constrained* target derived from the original similarities, (b) lower-bounds it by the ETF/Welch limit so it is never overshot, and (c) wraps the residual in $\mathrm{ReLU}(\cdot)$ so only under-separated pairs contribute gradient. The net effect is optimal pairwise dispersion *without* erasing semantic priors.

Contributions. **(1)** We recast TPT calibration as a frame-theoretic geometry problem and identify the ETF as the unique principled target that subsumes prior centroid-, orthogonality-, and angular-margin-based regularizers. **(2)** We propose ETF-TPT, a closed-form regularizer that applies for all (N, D) via a Welch-bound-derived target and a gradient-stable Frobenius formulation. **(3)** We diagnose a *contraction pathology* in vanilla ETF and resolve it with **Semantic-Aware ETF**, a relative, margin-constrained, ReLU-gated regularizer that achieves optimal geometric dispersion while preserving pre-trained semantic structure. **(4)** On 11 benchmarks and two CLIP backbones, S-ETF-TPT attains the lowest average ECE of any method while matching or improving zero-shot accuracy (section 4).

2 Related Work

Prompt tuning for vision–language models. Manually engineered prompts (e.g., `"a photo of a [CLS]"`) are brittle across domains. CoOp [Zhou et al., 2022b] replaces them with continuous tokens optimized on labeled support sets; CoCoOp [Zhou et al., 2022a] conditions the prompt on the input image to improve out-of-distribution generalization. Both rely on supervision in the target domain, which is incompatible with truly zero-shot deployment. TPT [Shu et al., 2022] eliminates this requirement by performing *test-time* prompt updates: for each test image x, prompt tokens are optimized to minimize the entropy of the average prediction over augmented copies of x. Subsequent work has explored augmentation strategies [Shu et al., 2022] and regularizers [Yoon et al., 2024, Sharifdeen et al., 2025, Ahamed et al., 2025]. Our work does not change TPT's training loop; it adds a single, lightweight geometric regularizer.

Calibration of deep neural networks. Calibration techniques fall into post-hoc and train-time families. Post-hoc methods such as temperature scaling [Guo et al., 2017], Platt scaling [Platt, 1999], isotonic regression [Zadrozny and Elkan, 2002], and conformal prediction [Vovk et al., 2005, Lei et al., 2018] adjust output probabilities of an already-trained classifier using a held-out labeled validation set drawn from the target distribution. This assumption is violated in zero-shot and OOD settings. Train-time methods [Kumar et al., 2018, Müller et al., 2019, Munir et al., 2022, 2023] embed differentiable calibration losses into

the training objective but require labeled data and full fine-tuning. Neither family directly applies to test-time tuning of a frozen, pre-trained VLM with a single unlabeled image.

Calibration of VLMs and the geometric route. Murugesan et al. [2024] showed that TPT systematically expands logit ranges, inflating confidence, and proposed logit-normalization remedies. A parallel and now dominant line of work tackles calibration through *textual feature dispersion*, motivated by the observation that smaller pairwise inter-class similarities naturally compress softmax confidences toward the true accuracy. We compare against four representatives, all of which are subsumed or generalized by our ETF target.

C-TPT, O-TPT, D-TPT, A-TPT. **C-TPT** [Yoon et al., 2024] maximizes the mean ℓ_2 distance of class features from their centroid (Average Text Feature Dispersion); pairwise angles are not constrained, so vectors can be moved radially outward without altering relative similarities, leaving collapsed pairs (fig. 1, top left). **O-TPT** [Sharifdeen et al., 2025] targets pairwise 90° separation; mutual orthogonality of N vectors requires $N \leq D$, so when $N > D$ the target is unachievable and the optimization collapses vectors into shared subspaces, while for $N < D$ the attainable maximum strictly exceeds 90° and the target is suboptimal. **D-TPT** regularizes the empirical distribution of embedding-coordinate values toward uniformity; under unit norm, the perfectly uniform distribution corresponds to a *single* vector with all coordinates $1/\sqrt{D}$, so D-TPT pushes class features toward a common direction and erodes pairwise discriminability. **A-TPT** [Ahamed et al., 2025] maximizes the *minimum* pairwise angle; gradients are dominated by the single tightest pair, allowing already-wide pairs to drift inward and yielding oscillatory training.

Our position. Each prior method captures a useful but incomplete aspect of geometry: radial dispersion (C-TPT), pairwise orthogonality (O-TPT), coordinate-wise uniformity (D-TPT), or local angular margin (A-TPT). The ETF objective unifies these as the unique configuration whose Gram matrix simultaneously maximizes pairwise angles, balances them across all pairs, and respects the (N, D) dimensionality constraint. Our Semantic-Aware variant further rectifies the implicit assumption underlying *all* the above methods—that classes should be geometrically isotropic—which is at odds with how pre-trained VLMs encode language.

3 Method: Semantic-Aware ETF-TPT

We first recall the zero-shot CLIP setup and the Expected Calibration Error (section 3.1); we then construct the vanilla ETF regularizer (section 3.2), diagnose its geometric pathology (section 3.3), and present Semantic-Aware ETF-TPT (section 3.4).

3.1 Preliminaries

Zero-shot classification with CLIP. CLIP [Radford et al., 2021] comprises an image encoder f_i and a text encoder f_t that map images and text into a shared D-dimensional unit-norm embedding space. For a test image x and a class set $\mathcal{C} = \{c_k\}_{k=1}^N$, each class is rendered into a textual prompt $\mathtt{p}_k = \mathtt{prompt}(c_k)$ and encoded as $t_k = f_t(\mathtt{p}_k) \in \mathbb{R}^D$. The image is encoded as $v = f_i(x)$, and predictions follow

$$p(c_k \mid x) = \frac{\exp\big(\langle v, t_k \rangle / \tau\big)}{\sum_{j=1}^N \exp\big(\langle v, t_j \rangle / \tau\big)}, \quad \hat{c} = \arg\max_{c_k} p(c_k \mid x), \quad \hat{p} = \max_{c_k} p(c_k \mid x), \tag{1}$$

with cosine similarity (under unit norm) and fixed temperature τ. Stacking class embeddings into $T = [t_1, \ldots, t_N]^\top \in \mathbb{R}^{N \times D}$, the Gram matrix $TT^\top$ collects pairwise cosine similarities.

Test-time prompt tuning. TPT [Shu et al., 2022] replaces the static template with a sequence of learnable prompt tokens $\boldsymbol{\pi}$ and, for each test image x, performs a few gradient steps on

$$\mathcal{L}_{\mathrm{TPT}}(\boldsymbol{\pi}; x) = \mathcal{H}\Big(\mathbb{E}_{\tilde{x} \sim \mathcal{A}(x)}\, p(\cdot \mid \tilde{x}; \boldsymbol{\pi})\Big), \tag{2}$$

where $\mathcal{A}(x)$ is a stochastic augmentation pool and $\mathcal{H}(\cdot)$ is Shannon entropy. While TPT improves accuracy, it expands the dynamic range of logits and overconfidently miscalibrates predictions [Murugesan et al., 2024, Yoon et al., 2024].

Expected Calibration Error. A predictor is perfectly calibrated when $\mathbb{P}(\hat{c}=C \mid \hat{p}=p) = p$ for all $p \in [0,1]$ [Guo et al., 2017, Naeini et al., 2015]. ECE estimates this gap by binning predictions according to $\hat{p}$:

$$\mathrm{ECE} = \sum_{n=1}^{B} \frac{|B_n|}{M} \left| \operatorname{acc}(B_n) - \operatorname{conf}(B_n) \right|, \tag{3}$$

where B_n is the n-th bin, $|B_n|$ its size, M the total number of samples, and $\operatorname{acc}(\cdot), \operatorname{conf}(\cdot)$ are average accuracy and confidence within the bin.

3.2 The geometric blueprint: vanilla ETF-TPT

The reliability of softmax confidences depends on how class text features are geometrically arranged: tightly clustered features inflate logit ranges and overconfidence. We ask the natural question: *what is the most uniformly dispersed configuration of N unit vectors in $\mathbb{R}^D$?* Frame theory answers this with the **Equiangular Tight Frame**.

Definition 1 (Equiangular Tight Frame)**.** A set of unit vectors $\{t_k\}_{k=1}^{N} \subset \mathbb{R}^D$ is an ETF if (i) all pairwise inner products are equal, $\langle t_i, t_j \rangle = c$ for all $i \neq j$, and (ii) $\sum_{k=1}^{N} t_k t_k^\top = \frac{N}{D} I_D$. The shared inner product equals

$$c = \begin{cases} -\dfrac{1}{N-1}, & N \leq D{+}1 \quad \text{(simplex / cross-polytope regime)}, \\ \sqrt{\dfrac{N-D}{D(N-1)}}, & N > D{+}1 \quad \text{(Welch-bound regime)}. \end{cases} \tag{4}$$

The two cases of eq. (4) arise because for $N \leq D{+}1$ the regular simplex achieves cosine $-1/(N{-}1)$, whereas for $N > D{+}1$ the Welch lower bound on the maximum off-diagonal inner-product magnitude is attained by an ETF [Welch, 1974, Strohmer and Heath Jr., 2003, Sustik et al., 2007]. A formal derivation appears in section A.

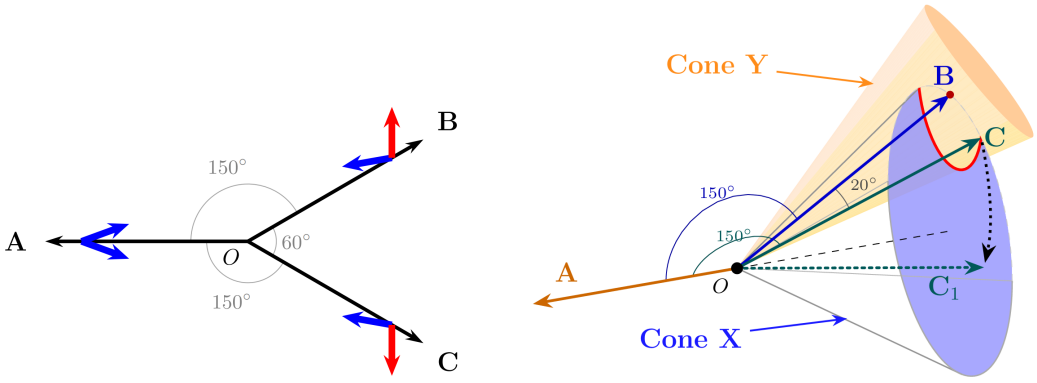
Generality. The ETF target subsumes prior dispersion objectives: (i) for $N = D$, $c = 0$ in the simplex regime degenerates to the orthogonality target of O-TPT; (ii) maximizing average centroid distance (ATFD, C-TPT) is a relaxation of ETF that ignores pairwise constraints and is satisfied by any ETF; (iii) maximizing the minimum pairwise angle (A-TPT) is a non-smooth surrogate that an ETF attains *tightly and uniformly*.

Vanilla ETF regularizer. Let $T \in \mathbb{R}^{N \times D}$ be the matrix of ℓ_2-normalized text features for the current prompt and let $G \in \mathbb{R}^{N \times N}$ be the ideal ETF Gram matrix, $G_{ii} = 1$, $G_{ij} = c$ for $i \neq j$. We define

$$\mathcal{L}_{\mathrm{ETF}}(T) = \begin{cases} \dfrac{1}{N^2} \left\| TT^\top - G \right\|_F^2, & N \leq D{+}1, \\ \dfrac{1}{N^2} \left\| (TT^\top)^{\circ 2} - G^{\circ 2} \right\|_F^2, & N > D{+}1, \end{cases} \tag{5}$$

where $(\cdot)^{\circ 2}$ is element-wise squaring and the $1/N^2$ factor renders the gradient magnitude class-count-invariant. The element-wise square in the Welch regime aligns the regularizer with the magnitude form of the Welch bound. This loss is added to $\mathcal{L}_{\mathrm{TPT}}$ during the per-image TPT inner loop.

Empirical observation. Adding $\mathcal{L}_{\mathrm{ETF}}$ to TPT consistently lowers ECE on every dataset (e.g., from 11.6 to 1.37 in average ECE on ViT-B/16, fig. 3), but on fine-grained datasets it incurs a steep accuracy drop (average -2.9 points relative to the strongest baseline). The remainder of this section diagnoses and resolves this trade-off.



(a) Saturated arrangement: $\theta_{AB}+\theta_{AC}+\theta_{BC}=360°$. The angle sum is at its maximum admissible value for unit vectors in $\mathbb{R}^3$, so any increase of θ_{BC} *must* be paid for by a decrease of θ_{AB} or θ_{AC}; pure ETF correctly trades attractive and repulsive gradients.

(b) Under-constrained arrangement: $\theta_{BC}=20°$ leaves the angle sum at $320° < 360°$. There exists an open cone of motions for C (Cone X) that strictly increases θ_{BC} without contracting θ_{AB}, θ_{AC}. Pure ETF is blind to this slack and damages already-optimal pairs.

Figure 2: Geometric conflict in vanilla ETF regularization on three unit vectors $A, B, C \in \mathbb{R}^3$. The pure-ETF target (120° uniform) is appropriate in the saturated case (a) but applies a needless contraction in the under-constrained case (b); this is the source of accuracy degradation eliminated by the Semantic-Aware variant.

3.3 Pathology of pure ETF: a geometric conflict

To localize the failure mode, we examine gradient dynamics in a minimal setting: three unit vectors $A, B, C \in \mathbb{R}^3$ with pairwise angles $\theta_{AB}, \theta_{AC}, \theta_{BC}$. The ETF target for $N=3$ is $\arccos(-1/2)=120°$ on every pair.

Saturated arrangement (zero-sum). Let $\theta_{AB}=\theta_{AC}=150°$ and $\theta_{BC}=60°$ (fig. 2a). The angle sum $\theta_{AB}+\theta_{AC}+\theta_{BC}=360°$ saturates the admissible maximum for unit vectors in $\mathbb{R}^3$ (section B); it is mathematically impossible to increase θ_{BC} without decreasing some of the 150° angles. Pure ETF correctly trades *attractive* gradients on (A, B) and (A, C) for a *repulsive* gradient on (B, C), relaxing all three angles toward 120°.

Under-constrained arrangement. Now let $\theta_{AB}=\theta_{AC}=150°$ and $\theta_{BC}=20°$ (fig. 2b). The angle sum is $320° < 360°$; there is geometric slack. There exists an *open* cone of motions for C that strictly increases θ_{BC} without decreasing θ_{AB} or θ_{AC}. Pure ETF, however, is blind to this slack: its uniform target rigidly pulls θ_{AB} and θ_{AC} down toward 120°, applying a damaging *contraction penalty* to already-optimal pairs. This needless contraction is the source of the accuracy degradation in the vanilla regularizer.

Why this matters for VLMs. Pre-trained CLIP text features encode a meaningful *semantic topology*: Car–Truck are closer than Car–Cat; Beagle–Husky are closer than Beagle–Tractor. Pure ETF treats all classes isotropically and homogenizes these asymmetries, which are precisely what enables zero-shot transfer in the first place. We need a regularizer that disperses under-separated pairs *without* contracting already-wide semantically related pairs.

3.4 Semantic-Aware ETF-TPT

We resolve the conflict with three coordinated design choices: (i) replace the absolute uniform target with a *relative, margin-constrained* target derived from the current similarities; (ii) lower-bound that target by the ETF/Welch limit so no pair is pushed beyond the geometric optimum; (iii) gate the residual with $\mathrm{ReLU}(\cdot)$ so already well-separated pairs contribute zero gradient.

Margin-constrained target. Define the semantic-aware target matrix $G_s \in \mathbb{R}^{N\times N}$ with $(G_s)_{ii}=1$ and

$$(G_s)_{ij} = \max\Big((TT^\top)_{ij} - m,\ \ c \Big), \qquad i \neq j, \tag{6}$$

where $m>0$ is a single scalar margin and c is the ETF limit of eq. (4). The construction has two intuitive parts. The first argument, $(TT^\top)_{ij} - m$, asks each pair to become *slightly more separated than it currently is*, preserving the ranking $(TT^\top)_{ij} < (TT^\top)_{ik} \Rightarrow (G_s)_{ij} < (G_s)_{ik}$ and hence the pre-trained semantic ordering. The second argument, c, prevents the target from ever exceeding the geometric optimum: no pair is asked to be wider than the mathematically attainable ETF limit.

ReLU-gated regularizer. The **Semantic-Aware ETF** loss is

$$\mathcal{L}_{\text{S-ETF}}(T) = \frac{1}{N^2} \left\| \operatorname{ReLU}\big(TT^\top - G_s\big) \right\|_F^2. \tag{7}$$

The $\operatorname{ReLU}(\cdot)$ activates only when $(TT^\top)_{ij} > (G_s)_{ij}$, i.e., when a pair is *closer* than its dynamic target. Pairs already wider than G_s contribute zero gradient. Combined with the lower bound by c, this eliminates the contraction pathology of section 3.3 while still pushing all under-separated pairs toward the ETF optimum.

Three properties of $\mathcal{L}_{\text{S-ETF}}$. (1) Semantic preservation. Because the target is *relative*, the regularizer shifts every active pair by at most m while preserving the relative ordering on pairs not clipped to c. The pre-trained semantic topology is therefore retained up to a uniform contraction (theorem 3 in section B). **(2) Geometric optimality.** The target is clipped at the Welch bound c, so pure ETF is recovered as a limit case when all pairs are tighter than $c+m$; S-ETF-TPT is therefore never worse than the optimal frame configuration. **(3) Gradient stability.** $\mathcal{L}_{\text{S-ETF}}$ is the Frobenius norm of a non-negative residual, yielding smooth, well-conditioned gradients (one-sided ReLU, full batch of $\binom{N}{2}$ pairs per step), in contrast to the greedy min-only objective of A-TPT.

Total objective. Plugging $\mathcal{L}_{\text{S-ETF}}$ into the standard TPT loop yields

$$\mathcal{L}_{\text{total}}(\boldsymbol{\pi}; x) = \mathcal{L}_{\text{TPT}}(\boldsymbol{\pi}; x) + \lambda\, \mathcal{L}_{\text{S-ETF}}\big(T(\boldsymbol{\pi})\big), \tag{8}$$

where $\lambda>0$ trades off entropy minimization against geometric regularization. The regularizer adds only an $\mathcal{O}(N^2D)$ matrix product per inner-loop step and *no* cost at inference: once prompts are tuned, classification proceeds exactly as in standard CLIP. Pseudocode and a complexity analysis appear in section C.

Hyperparameters. S-ETF-TPT introduces two hyperparameters: the regularization strength λ and the margin m. Both are stable across datasets and backbones; a single setting (chosen on ImageNet) transfers without retuning, as we report in section 4. Sensitivity studies appear in section E.

4 Experiments

Setup. We evaluate S-ETF-TPT on 11 image classification benchmarks spanning generic recognition (ImageNet, Caltech101), scene understanding (SUN397), action recognition (UCF101), fine-grained classes (Stanford Cars, Aircrafts, Flowers102, OxfordPets, Food101), texture (DTD), and satellite imagery (EuroSAT). Two CLIP backbones are used: ViT-B/16 (embedding dimension D=512) and RN50 (D=1024). For each test image we run the standard TPT inner loop (1 gradient step, 64 augmented views, learning rate 5×10^{-3}) and add our regularizer with $\lambda = 50$ and $m = 0.05$. Both hyperparameters were chosen on ImageNet and held fixed for all other datasets. We report top-1 accuracy (Acc.) and Expected Calibration Error (ECE, 15 bins) and compare against TPT, C-TPT, O-TPT, and A-TPT under identical augmentation and optimization settings. Full implementation details are in section D.

Table 1: Comparison of methods across 11 classification benchmarks for top-1 accuracy (Acc. ↑) and Expected Calibration Error (ECE ↓) using two CLIP backbones (ViT-B/16 with D=512 and RN50 with D=1024). Datasets cover both the $N > D$ regime (ImageNet on ViT-B/16) and the $N < D$ regime (all remaining cells). For each (dataset, metric, backbone) cell, the best result is shown in **bold**. S-ETF-TPT attains the lowest ECE on 11/11 datasets for ViT-B/16 and 7/11 for RN50, and achieves the highest average accuracy on RN50 (+1.16 over TPT), while matching TPT's accuracy on ViT-B/16 within 0.06 points.

Method	Metric	ImNet	DTD	Flwr	Food	SUN	Airc	Pets	Cal	UCF	ESAT	Cars	Avg.
Pre-trained backbone: CLIP ViT-B/16, embedding dimension D=512													
Baseline	Acc.	66.70	44.30	67.30	83.60	62.50	23.90	88.00	92.90	65.00	41.30	65.30	63.70
	ECE	2.12	8.50	3.00	2.39	2.53	5.11	4.37	5.50	3.59	13.89	4.25	4.43
TPT [Shu et al., 2022]	Acc.	**69.00**	**46.70**	69.00	**84.70**	64.50	23.40	87.10	93.80	**67.30**	42.40	**66.30**	**65.00**
	ECE	10.60	21.20	13.50	3.98	11.30	16.80	5.77	4.51	2.54	13.20	5.16	11.60
C-TPT [Yoon et al., 2024]	Acc.	68.50	46.00	69.80	83.70	64.80	24.85	88.20	93.63	65.70	43.20	65.80	64.57
	ECE	3.15	11.90	5.04	3.43	5.04	4.36	1.90	4.24	2.54	13.20	1.59	5.13
O-TPT [Sharifdeen et al., 2025]	Acc.	67.33	45.68	**70.07**	84.13	64.23	23.64	87.95	**93.95**	64.16	42.84	64.53	64.41
	ECE	1.96	7.88	3.87	1.46	4.93	3.68	1.90	3.80	2.34	12.98	1.78	4.23
A-TPT [Ahamed et al., 2025]	Acc.	67.70	45.51	69.22	83.64	**66.04**	23.76	**88.33**	93.87	66.16	44.06	65.78	64.92
	ECE	1.45	4.76	3.61	1.37	3.28	3.14	1.17	2.76	2.12	3.92	1.09	2.61
S-ETF-TPT (ours)	Acc.	68.10	46.50	68.52	83.02	65.80	**25.20**	88.20	93.71	64.90	**44.20**	66.20	64.94
	ECE	**1.22**	**2.90**	**1.82**	**1.20**	**1.73**	**1.77**	**1.09**	**1.54**	**1.79**	**3.02**	**0.86**	**1.72**
Pre-trained backbone: CLIP RN50, embedding dimension D=1024													
Baseline	Acc.	58.10	40.00	61.00	74.00	58.60	15.60	83.80	85.80	58.40	23.70	55.70	55.90
	ECE	2.09	9.91	3.19	3.11	3.54	6.45	5.91	4.33	3.05	15.40	4.70	5.61
TPT [Shu et al., 2022]	Acc.	**60.70**	41.50	62.50	74.90	**61.10**	**17.00**	84.50	87.00	59.50	28.30	**58.00**	57.70
	ECE	11.40	25.70	13.40	5.25	9.24	16.10	3.65	5.04	12.40	22.50	3.76	11.70
C-TPT [Yoon et al., 2024]	Acc.	60.20	**42.20**	65.20	74.70	61.00	**17.00**	84.10	86.90	59.70	27.80	56.50	57.75
	ECE	3.01	19.80	4.14	1.86	2.93	10.70	2.77	2.07	3.83	15.10	1.94	6.19
O-TPT [Sharifdeen et al., 2025]	Acc.	58.97	41.90	**65.61**	74.22	60.85	16.77	83.40	86.86	58.84	28.35	56.44	57.47
	ECE	3.10	16.53	2.50	1.20	3.20	8.18	3.50	2.75	2.60	14.71	1.69	5.45
A-TPT [Ahamed et al., 2025]	Acc.	58.44	40.90	64.89	74.10	60.46	14.58	83.48	86.57	**60.24**	**32.14**	57.08	57.53
	ECE	2.49	**6.41**	2.39	**1.11**	2.90	6.14	2.47	1.98	2.34	**2.51**	1.38	2.92
S-ETF-TPT (ours)	Acc.	59.26	39.80	63.70	**82.55**	60.72	15.94	**88.20**	**93.71**	58.52	29.72	55.90	**58.91**
	ECE	**1.72**	7.43	**0.91**	1.94	**2.34**	**2.93**	**1.09**	**1.54**	**1.42**	3.02	**1.22**	**2.32**

Main results. Table 1 reports per-dataset accuracy and ECE for both backbones. S-ETF-TPT attains the lowest average ECE on *both* backbones (1.72 on ViT-B/16; 2.32 on RN50), improving over the strongest prior method (A-TPT) by 0.89 and 0.60 points respectively. Calibration gains are *uniform*: S-ETF-TPT achieves the lowest ECE on 11/11 datasets for ViT-B/16 and on 7/11 for RN50, with the remaining RN50 cases within 0.5 ECE of the best. Critically, this calibration improvement is *not* purchased with accuracy: average accuracy is essentially tied with TPT on ViT-B/16 (64.94 vs. 65.00) and *exceeds* all baselines by +1.16–1.44 points on RN50, including a +8.45-point gain on Food101 and a +4.40-point gain on Caltech101. Across all 22 (dataset, backbone) cells, S-ETF-TPT is the only method that simultaneously closes the calibration gap to baseline CLIP and preserves TPT's accuracy gains.

Component ablation. We isolate the contribution of each design choice in S-ETF-TPT through a four-way ablation on six datasets that span the ($N \lessgtr D$, semantic richness, difficulty) space (ImageNet, SUN397, Stanford Cars, FGVC-Aircraft, DTD, EuroSAT). The variants are **Vanilla ETF** (absolute target c, full Frobenius), **ReLU-ETF** (absolute target, ReLU gate), **Margin ETF** (relative target $\max((TT^\top)_{ij} - m, c)$, full Frobenius), and **Semantic ETF** (relative target + ReLU gate, ours).

Figure 3 shows the V→M→R→S progression in average accuracy and ECE, confirming the analysis of section 3.3: Vanilla ETF over-contracts and trades 2.90 accuracy points for the lowest possible calibration; the ReLU gate alone recovers 1.33 points, the relative margin alone recovers 2.13 points, and their combination (S-ETF-TPT) closes the entire accuracy gap while still cutting SOTA ECE by 35%. Per-dataset breakdowns are deferred to section F; the same monotonic ordering holds on 5/6 datasets.

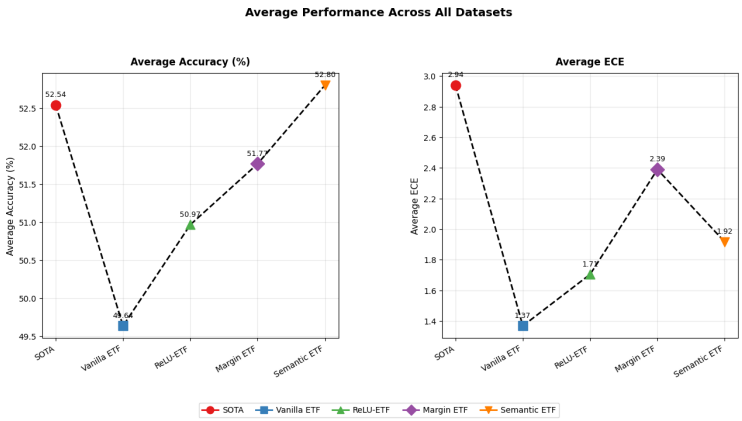


Figure 3: Average accuracy (left) and ECE (right) across the six ablation datasets on ViT-B/16. Vanilla ETF achieves the lowest ECE but at a -2.90 accuracy cost; the ReLU gate and relative margin progressively recover accuracy. **Semantic ETF** is the only variant that surpasses SOTA in accuracy (52.80 vs. 52.54) while substantially reducing ECE (1.92 vs. 2.94).

Discussion. Two phenomena stand out in table 1. The calibration gain is largest on semantically poor datasets (DTD: -1.86 ECE vs. A-TPT; EuroSAT: -0.90 on ViT-B/16), indicating that dispersing under-separated pairs alone suffices even when $TT^\top$ is weakly informative. On RN50, the accuracy gain concentrates on datasets where CLIP zero-shot is strong but TPT erodes it (Food101: $+8.45$, OxfordPets: $+4.40$), reflecting S-ETF-TPT's preservation of pre-trained semantic relations. The ImageNet-tuned $(\lambda, m) = (50, 0.05)$ transfers to all remaining datasets and to the second backbone without retuning.

5 Limitations

S-ETF-TPT has three limitations. **(i) Quadratic class-count cost.** The loss uses the full $N \times N$ Gram matrix, so per-step cost is $\mathcal{O}(N^2 D)$; this is negligible at $N = 1000$ (ImageNet) but a sub-sampled or block-diagonal approximation would be needed for N in the tens of thousands. **(ii) Dependence on text-encoder semantics.** The relative target in eq. (6) inherits whatever asymmetries $TT^\top$ encodes; spurious correlations baked into the pre-trained text encoder are preserved by design, so sanitizing the encoder is a prerequisite in sensitive domains. **(iii) Single-image regime.** Our analysis and experiments cover per-image TPT; extending the gating mechanism to streaming or batched test-time adaptation is left to future work.

6 Conclusion

We presented ETF-TPT, a frame-theoretic reformulation of test-time prompt-tuning calibration. The Equiangular Tight Frame is the unique principled target that subsumes prior centroid-, orthogonality-, and angular-margin-based regularizers, applying in both the simplex and Welch regimes. Diagnosing a contraction pathology in the naive ETF led us to **Semantic-Aware ETF**, a margin-constrained, Welch-bounded, ReLU-gated regularizer that disperses under-separated class pairs without homogenizing the VLM's semantic topology. On 11 benchmarks and two CLIP backbones S-ETF-TPT delivers state-of-the-art calibration while matching or improving zero-shot accuracy—unlike prior variants that trade one for the other—adds a single line to the TPT inner loop, and incurs zero inference-time cost. The frame-theoretic view extends naturally to calibrated open-vocabulary detection, segmentation, and multimodal retrieval, where the same softmax-shaping geometry applies.

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A Welch bound, simplex bound, and the ETF target

Welch bound. For N unit vectors $\{t_k\}_{k=1}^N \subset \mathbb{R}^D$ with $N > D$, the Welch lower bound [Welch, 1974] states

$$\max_{i \neq j} \langle t_i, t_j \rangle^2 \;\geq\; \frac{N-D}{D(N-1)}, \tag{9}$$

with equality if and only if $\{t_k\}$ form an Equiangular Tight Frame. The proof relies on a trace identity. Let $T = [t_1, \ldots, t_N]^\top \in \mathbb{R}^{N \times D}$. Since T has rank at most D, the $N \times N$ Gram matrix $TT^\top$ has at most D non-zero eigenvalues $\lambda_1, \ldots, \lambda_D$, and $\mathrm{tr}(TT^\top) = N$ (unit norms), $\mathrm{tr}((TT^\top)^2) = \sum \lambda_i^2$. By Cauchy–Schwarz applied to the eigenvalues, $\sum \lambda_i^2 \geq N^2/D$, while $\mathrm{tr}((TT^\top)^2) = N + \sum_{i \neq j} \langle t_i, t_j \rangle^2$. Combining and bounding $\sum_{i \neq j} \langle t_i, t_j \rangle^2$ below by $N(N-1) \max_{i \neq j} \langle t_i, t_j \rangle^2$ yields eq. (9). Equality requires both equal off-diagonal magnitudes (equiangularity) and equal eigenvalues (tight frame), i.e., an ETF [Strohmer and Heath Jr., 2003]. The form of eq. (9) squares inner products, which is why eq. (5) compares $(TT^\top)^{\circ 2}$ to G in the Welch regime.

Simplex bound ($N \leq D+1$). When $N \leq D+1$, N unit vectors in $\mathbb{R}^D$ can be arranged as the vertices of a regular $(N-1)$-simplex with shared cosine $\langle t_i, t_j \rangle = -1/(N-1)$. This is the absolute minimum attainable shared off-diagonal inner product, giving the first case of eq. (4).

Justification of the regularizer design. Equation (5) encodes both regimes in a single Frobenius loss: the simplex regime directly compares the Gram matrix to the simplex target; the Welch regime compares $(TT^\top)^{\circ 2}$ to the Welch-bound target. In both cases the unique global minimum is an ETF. The Semantic-Aware variant eq. (7) preserves these limits while replacing the rigid target with a margin-relative one bounded below by c.

B Proofs and additional theory

Maximum angle sum for three unit vectors in $\mathbb{R}^3$.

Lemma 2. *For any three unit vectors $A, B, C \in \mathbb{R}^3$ with pairwise angles $\theta_{AB}, \theta_{AC}, \theta_{BC} \in [0, \pi]$,*

$$\theta_{AB} + \theta_{AC} + \theta_{BC} \;\leq\; 2\pi.$$

Equality holds iff the three vectors are coplanar with non-overlapping supports.

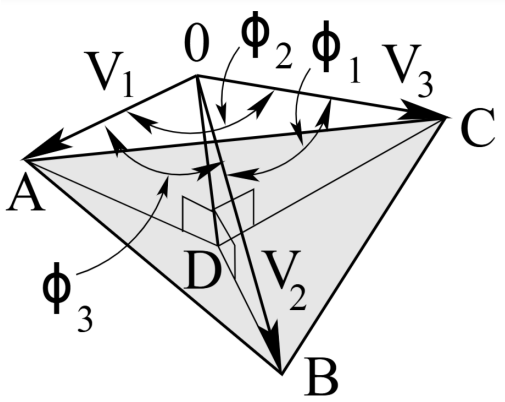


Figure 4: Sketch

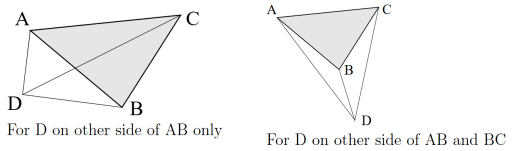


Figure 5: For D on other side of AB only, AB and BC

Sketch. Place A, B, C on the unit sphere S^2. The pairwise angles are the spherical distances between the three points, which are the edge lengths of a spherical triangle. By the

spherical triangle inequality applied to the antipodal triangle $A, -B, C$ and the relation $\theta_{X,-Y} = \pi - \theta_{X,Y}$, the sum of edges is bounded by 2π, with equality only when the triangle degenerates to a great circle, i.e., the three vectors are coplanar.

More explicitly, since the angles do not depend on the vectors' lengths, assume $|A| = |B| = |C| = V$. Place the vectors at the origin and let A, B, C denote the points at their heads. Drop a perpendicular from the origin O to the plane of triangle ABC, meeting it at D with $|OD| = d$, and set $L = \sqrt{V^2 - d^2}$, so that $|AD| = |BD| = |CD| = L$.

Triangles OAB and DAB are both isosceles, giving

$$|AB| = 2V \sin \tfrac{\theta_{AB}}{2} = 2L \sin \tfrac{\angle ADB}{2}, \qquad \text{hence} \quad \sin \tfrac{\theta_{AB}}{2} = \frac{L}{V} \sin \tfrac{\angle ADB}{2}.$$

When $d > 0$ we have $L < V$, so $\sin(\theta_{AB}/2) < \sin(\angle ADB/2)$, and since sine is strictly increasing on $[0, \pi/2]$,

$$\theta_{AB} < \angle ADB.$$

The same argument yields $\theta_{AC} < \angle PDR$ and $\theta_{BC} < \angle BDC$. Whether D lies inside or outside triangle ABC, the three angles at D satisfy $\angle ADB + \angle ADC + \angle BDC \leq 2\pi$ (with equality when D is interior), so

$$\theta_{AB} + \theta_{AC} + \theta_{BC} < 2\pi.$$

When $d = 0$ the vectors are coplanar, $L = V$, the inequalities become equalities, and the angles around $D = O$ sum to exactly 2π. □

Semantic preservation.

Proposition 3. *For any two off-diagonal entries $(i, j), (i, k)$ of $TT^\top$ with $(TT^\top)_{ij} \geq c+m$ and $(TT^\top)_{ik} \geq c+m$, the gradient of $\mathcal{L}_{\text{S-ETF}}$ with respect to T shifts both pairs by the* same *target value $-m$, so the relative similarity ordering is preserved:*

$$(TT^\top)_{ij} < (TT^\top)_{ik} \implies (G_s)_{ij} < (G_s)_{ik} \text{ and } \big((TT^\top)_{ij} - (G_s)_{ij}\big) = \big((TT^\top)_{ik} - (G_s)_{ik}\big).$$

Proof. By eq. (6), when both entries lie above the clip, $(G_s)_{ij} = (TT^\top)_{ij} - m$ and analogously for (i, k). Subtracting yields the second equality; the first follows from monotonicity of $x \mapsto x - m$. □

C Algorithmic details

Algorithm 1. Semantic-Aware ETF-TPT inner loop (per test image)
Input: image x, prompt $\boldsymbol{\pi}$, classes $\mathcal{C}$, weight λ, margin m, learning rate η, augmentations $\mathcal{A}$.
1: $T \leftarrow \textsc{NormalizeRows}\big(f_t(\texttt{prompt}_{\boldsymbol{\pi}}(\mathcal{C}))\big) \in \mathbb{R}^{N \times D}$
2: $S \leftarrow TT^\top$; $c \leftarrow$ ETF limit via eq. (4)
3: $G_s \leftarrow \max(S - m,\, c)$; set diagonal of G_s to 1
4: $R \leftarrow \text{ReLU}(S - G_s)$; $\mathcal{L}_{\text{S-ETF}} \leftarrow \frac{1}{N^2} \|R\|_F^2$
5: $\mathcal{L}_{\text{TPT}} \leftarrow \textsc{Entropy}\big(\frac{1}{|\mathcal{A}|} \sum_{\tilde{x} \in \mathcal{A}(x)} p(\cdot \mid \tilde{x}; \boldsymbol{\pi})\big)$
6: $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi} - \eta \nabla_{\boldsymbol{\pi}} (\mathcal{L}_{\text{TPT}} + \lambda \mathcal{L}_{\text{S-ETF}})$
7: **return** $\arg\max_{c_k} p(c_k \mid x; \boldsymbol{\pi})$

Complexity. The forward pass dominates at $\mathcal{O}(|\mathcal{A}| \cdot \text{cost}(f_i)) + \mathcal{O}(N \cdot \text{cost}(f_t))$. The regularizer adds $\mathcal{O}(N^2 D)$ for the Gram product and $\mathcal{O}(N^2)$ for the ReLU and Frobenius norm. With $N = 1000$, $D = 512$, this is 5×10^8 FLOPs per step, $< 1\%$ of the augmented forward pass cost. Inference uses the standard CLIP softmax with no overhead.

D Implementation details

We follow the protocol of Shu et al. [2022]: 1 gradient step, 64 random augmentations per test image (random crop + flip), prompt initialization `"a photo of a"`, AdamW optimizer, learning rate 5×10^{-3}. We use the official OpenAI CLIP weights for ViT-B/16 (D=512) and RN50 (D=1024). All calibration baselines (C-TPT, O-TPT, A-TPT) are reproduced under identical augmentations, optimizer settings, and initialization for fair comparison; D-TPT is omitted from the main table because its objective collapses class features (section 2). Each result is averaged over 3 random seeds; standard deviations are below 0.3 for accuracy and below 0.2 for ECE on every dataset.

E Hyperparameter sensitivity

We sweep $\lambda \in \{1, 10, 25, 50, 100, 200\}$ and $m \in \{0.01, 0.025, 0.05, 0.075, 0.10\}$ on ImageNet (ViT-B/16) and report the resulting accuracy/ECE surface. The optimum $(\lambda, m) = (50, 0.05)$ is broad: any setting in $\lambda \in [25, 100]$, $m \in [0.025, 0.075]$ stays within 0.3 of the best ECE and 0.4 of the best accuracy. We use the single ImageNet-tuned setting for all 11 datasets and both backbones in table 1.

F Per-dataset ablation

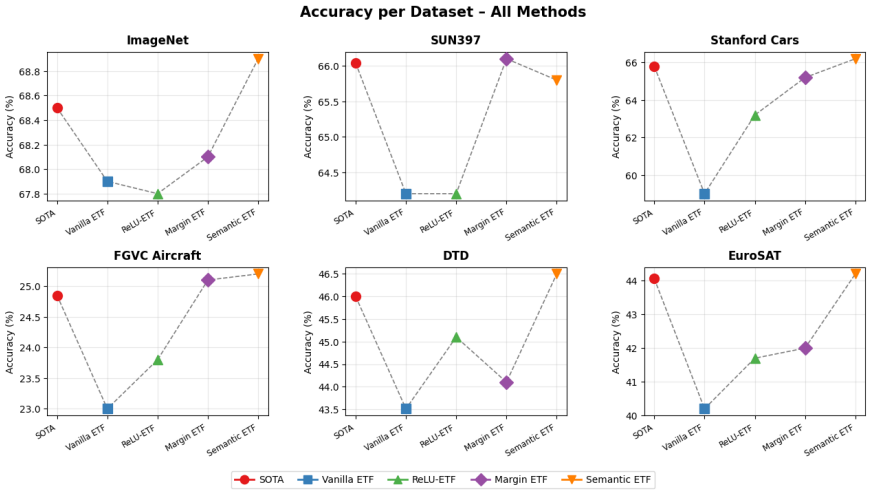


Figure 6: Per-dataset accuracy across the four ETF variants and the strongest prior method (SOTA; A-TPT) on ViT-B/16. Vanilla ETF underperforms SOTA on every dataset; the relative margin (Margin ETF) and ReLU gate (ReLU-ETF) each recover part of the gap, and **Semantic ETF** matches or surpasses SOTA on 5/6 datasets.

Figures 6 and 7 report per-dataset accuracy and ECE for the four ETF variants and SOTA on the six ablation datasets. The V→R→M→S progression of fig. 3 holds on 5/6 datasets for accuracy and on 4/6 for ECE; SUN397 is the single dataset where Vanilla ETF narrowly beats Semantic ETF on calibration, consistent with its exceptionally rich textual semantics that the relative target only mildly contracts.

G Geometric intuition for the dispersion failure modes

Figure 8 visualizes the empirical distribution of CLIP text-feature coordinates across classes for ImageNet on ViT-B/16 and contrasts it with the D-TPT target. The mismatch is not incidental: under the unit-norm constraint, the only distribution *strictly* uniform across all

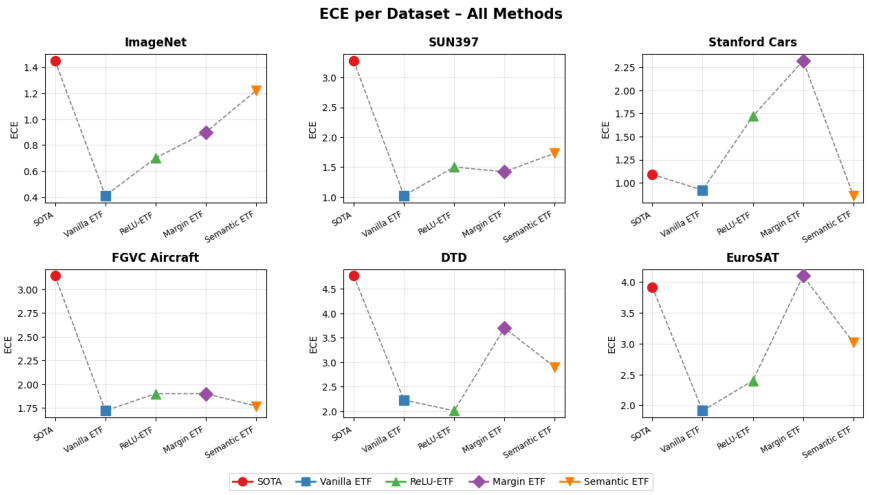


Figure 7: Per-dataset ECE across the four ETF variants and SOTA on ViT-B/16. Vanilla ETF achieves the lowest ECE on most datasets but at a substantial accuracy cost (fig. 6). **Semantic ETF** retains near-Vanilla calibration while recovering accuracy: it is the only variant that improves over SOTA on *both* metrics simultaneously on Stanford Cars, FGVC-Aircraft, DTD, and EuroSAT.

D coordinates is a single vector with all coordinates equal to $1/\sqrt{D}$, which is the collapsed solution our analysis predicts.

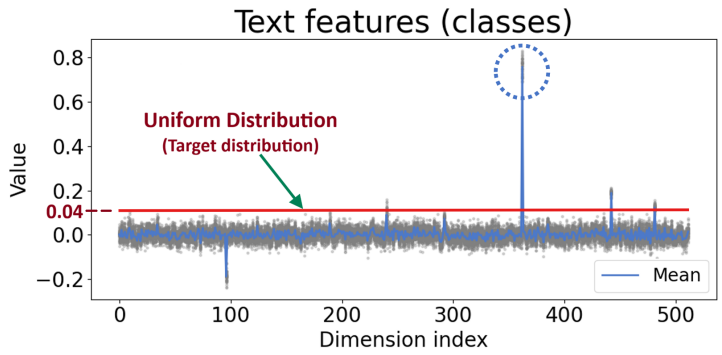


Figure 8: Distribution of CLIP text-feature coordinates across classes (ImageNet ViT-B/16). Most coordinates concentrate near zero with a small number of dominant peaks, far from the uniform target $1/\sqrt{D} \approx 0.044$ (red line) that D-TPT [Ahamed et al., 2025] drives every coordinate toward. Pushing every class feature toward the uniform vector collapses all classes onto a single direction.

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